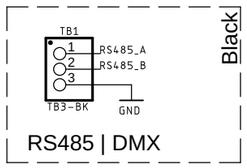
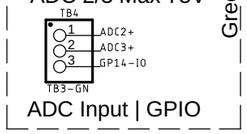


RS485 | DMX

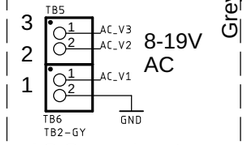
Black



ADC 2/3 Max 75V

ADC Input | GPIO

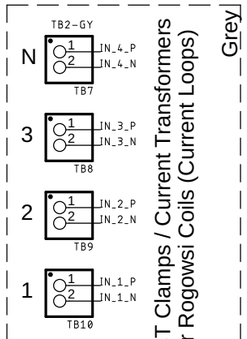
Green



8-19V AC

AC Sense Inputs

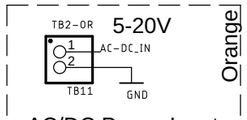
Grey



CT Clamps / Current Transformers or Rogowsi Coils (Current Loops)

CT Inputs

Grey

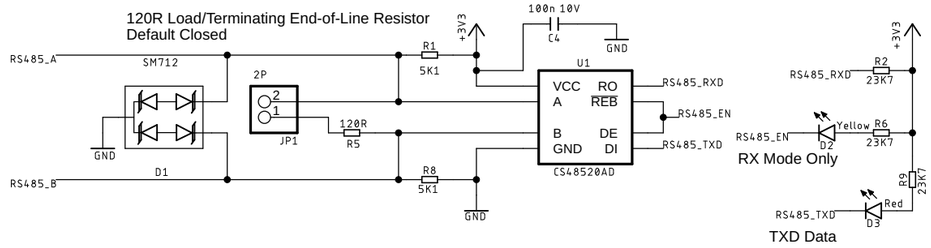


AC/DC Power Input

Orange

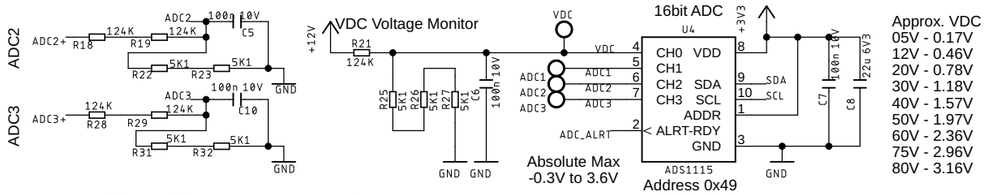
RS485 typically only uses A and B. Unless shielded cables are used.
DMX typically uses A, B with GND (Shield).
RS485 Ground is not isolated from Board Ground!

EN
0 = UART Driver Disabled and Receiver Enabled
1 = UART Driver Enabled (TX) = Default



RS-485 | Asymmetrical TVS | EFT | ESD | EOL Load Termination | MODBUS | DMX512

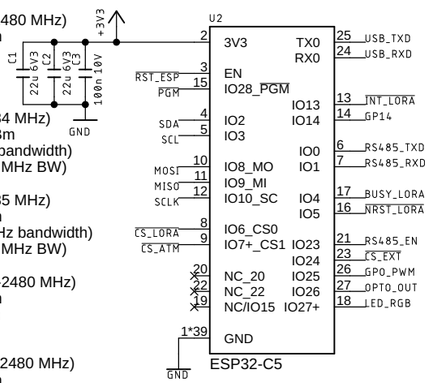
ADC2 and 3 will monitor 5-75V DC reference to Ground ADC 0 to 3.3V, with a resolution of 16 bits, resulting in a range of 0 to 4095 [I2C]



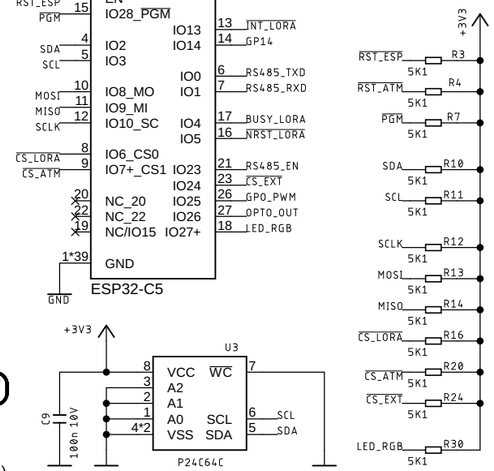
Approx. VDC
05V - 0.17V
12V - 0.46V
20V - 0.78V
30V - 1.18V
40V - 1.57V
50V - 1.97V
60V - 2.36V
75V - 2.96V
80V - 3.16V

Analog ADC | ADC Inputs | High Voltge Sensors

- Bluetooth (2402-2480 MHz)
2.4 GHz @ 20 dBm
Bluetooth LE
Bluetooth 5.3
Bluetooth Mesh
- Wi-Fi 6 (2412-2484 MHz)
2.4 GHz @ 19.5 dBm
802.11ax (20 MHz bandwidth)
802.11b/g/n (20/40 MHz BW)
- Wi-Fi 6 (5180-5885 MHz)
5 GHz @ 18.5 dBm
802.11ac/ax (20 MHz bandwidth)
802.11b/g/n (20/40 MHz BW)
- Thread 1.4 (2405-2480 MHz)
2.4 GHz @ 20 dBm
802.15.4 (2.4 GHz)
Thread Mesh
- Zigbee 3.0 (2405-2480 MHz)
2.4 GHz @ 20 dBm
802.15.4 (2.4 GHz)
Zigbee Mesh



~Mandatory Boot States
GP7 High
GP27 High
GP28 (PGM) High

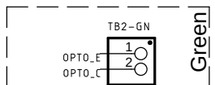


Variants:
ESP32-C5-WROOM-1-N8R8
ESP32-C5-WROOM-1U-N8R8 (U.FL)

NB. U.FL uses standard IPEX MHF1 Coax Connectors

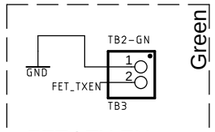
MCU | ESP32-C5 | EEPROM | 24C64

- 64K (8192 x 8)
- Examples Data Logging and Parameters Settings



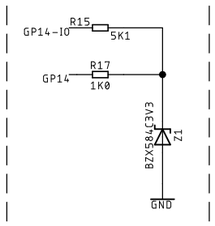
OPTO Output

Green



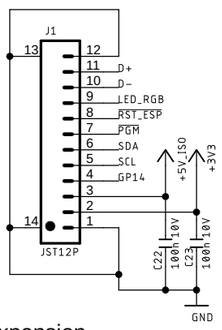
FET | TX EN

Green



GPIO14

DIN Control Panel

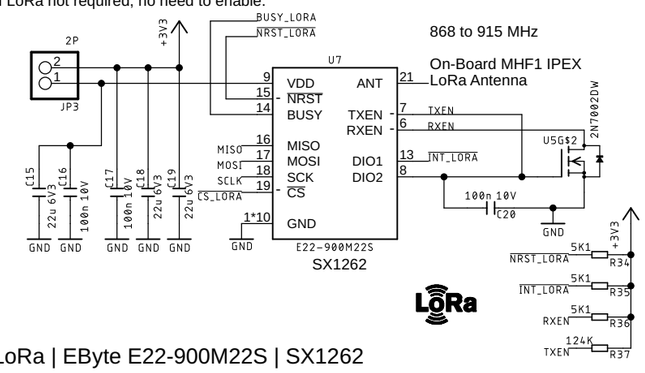


JST-12 SH VERT 1.0MM

Expansion

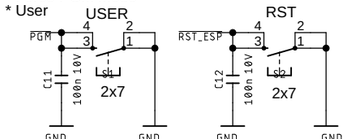
SX1262 LoRa 3V3 Enable
If LoRa not required, no need to enable.

[SPI]



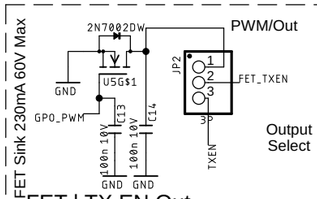
LoRa | EByte E22-900M22S | SX1262

User Button is Dual Purpose
* Force PGM Mode
* User

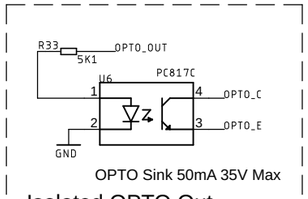


Tactile Switches

FET Sink 230mA 60V Max



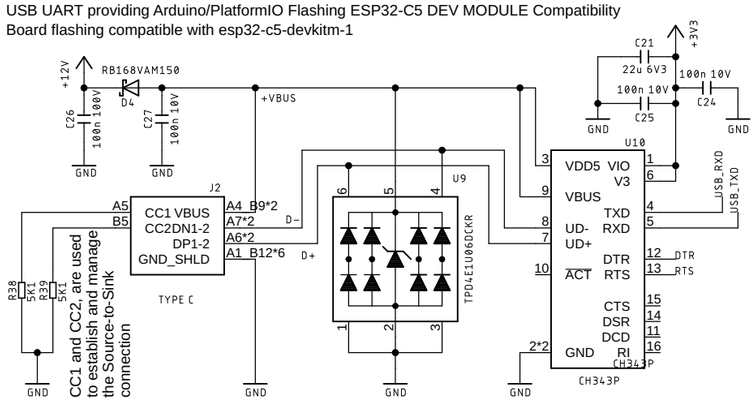
FET | TX EN Out



OPTO Sink 50mA 35V Max

Isolated OPTO Out

USB UART providing Arduino/PlatformIO Flashing ESP32-C5 DEV MODULE Compatibility
Board flashing compatible with esp32-c5-devkitm-1



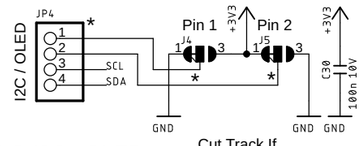
GPIO28 + Reset

Flash Boot	High
UART Download	Low

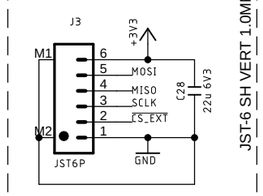
USB 1 | USB UART | Serial Monitoring | USB Flashing | Debug | ESD

Expansion Port (OLED/I2C) OLED Pin Configuration

Pin 1 - Default GND (1+2)
Pin 2 - Default +3V3 (1+2)



I/O I2C | OLED



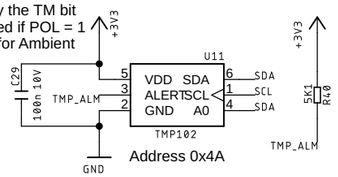
JST-6 SH VERT 1.0MM

SPI EXT

Alarm Alert

- Open Drain Output
- Default Low on Alert
- Alert pin is modified by the TM bit
- Polarity of bit is inverted if POL = 1
- Deduct approx. 10°C for Ambient

[I2C]



PCB Ambient Temperature | TMP102

IPEM Plus is an IoT Power Energy Monitor.
IPEM Plus is an ESP32-C5 SX1262 LoRa variant of the IPEM and EPEM series.

For further information and code examples, please see our code and GitHub WIKI pages. ditronix.net | GitHub.com/DitroniX | Hackster.io/DitroniX

Components optimised for BOM

DitroniX
Home Automation | Internet of Things | Energy Monitoring

Production

(C)2018-2026 Dave Williams | DitroniX | G8PUO

Enclosure VG-GR34 90x71x57
Main PCB 86 x 66.8

ESP32-C5 SX1262 SDK ditronix.net DitroniX Ltd – UK	IPEM Plus ESP32-C5 SX1262 E22-900M22S LoRa 1.2605.103	
	ATM90E36-ESP32-C5-LORA-IOT-POWER-ENERGY-MONITOR-BOARD-(PG1)	
	Sheet: 1/2	Drwg/Chk: Dave Williams
		06/05/2026 07:32

CT: Clamp Input - Based on YHDC SCT013-000 100A / 50mA 1:2000
RCT: Rogowski Current Transfomer e.g. PA3202NL

*Default
1/2 CT YDHC
2/3 Rogowski

Channel 1 (A) — Neg / Pos — IN_1_N / IN_1_P — JP5 / JP8 — JP6 / JP9 — CT1N / CT1P — R41 / R45 — R42 / R46 — C36 / C39 — RCT1N / RCT1P — ATM_I1_N / ATM_I1_P

Channel 2 (B) — Neg / Pos — IN_2_N / IN_2_P — JP10 / JP14 — JP11 / JP15 — CT2N / CT2P — R49 / R54 — R51 / R55 — C44 / C47 — RCT2N / RCT2P — ATM_I2_N / ATM_I2_P

Channel 3 (C) — Neg / Pos — IN_3_N / IN_3_P — JP16 / JP20 — JP17 / JP21 — CT3N / CT3P — R57 / R61 — R58 / R62 — C48 / C51 — RCT3N / RCT3P — ATM_I3_N / ATM_I3_P

Channel 4 (N) — Neg / Pos — IN_4_N / IN_4_P — JP24 / JP28 — JP25 / JP29 — CT4N / CT4P — R67 / R82 — R68 / R83 — C58 / C61 — RCT4N / RCT4P — ATM_I4_N / ATM_I4_P

3 x Phase Current RMS & 1 x Neutral Line Current RMS Monitoring

Sensor Inputs | CT Clamp | Rogowski

U13 — ATM90E36A-AU (3+1 Phase) Poly Phase Metering AFE

RESET, I1P, I1N, I2P, I2N, I3P, I3N, I4P(36), I4N(36), V1P, V1N, V2P, V2N, V3P, V3N, NC35, NC45, NC46, CS, SCLK, SDI, SDO, OSCO, OSCI, VDD1V8, DVDD, AVDD, IRQ0, IRQ1, ZX0, ZX1, ZX2, CF1, CF2, CF3, CF4, WARNOUT, PM0, PM1, VREF, DMA_CTRL(36), TEST, DGND, AGND

16.384 MHz — X1 — Guard Ring — GNDA

ATM_V1_P/N, ATM_V2_P/N, ATM_V3_P/N — R59 — C52 C53 C54 C55 C56 C57

Defaults
WO - L
ZXx - HL
IRQx - L
CFx - L

DMA Control

SPI Mode	*	L
DMA Mode		H

* Default Setting

Power Modes (PM1/PM0)

Normal (N) *	1+0 ON
Partial Measure	Only 0 ON
Detection (D)	Only 1 ON
Idle (I)	ALL OFF

ATM_WO - Warning Out
High = Error

CF1 - Active (Red)
CF2 - ReActive (Green)
CF3 - Fundamental (Blue)
CF4 - Harmonic (Amber/Ylw)
Pulse Outputs Status LEDs

DMA Control — JP7 — 5K1R44 — ATM_DMA

Power Modes (PM1 / PM0) — JP12 — 5K1R50 — ATM_PM0 — JP13 — 5K1R53 — ATM_PM1

SPLIT Phase ONLY = ON (e.g. USA)
Only Inputs 1 & 3 are used for SPLIT Phase

ATM_I4_N — JP18 — ATM_I4_P — JP19 — ATM_I2_N — JP22 — ATM_I2_P — JP23 — ATM_V2_N — JP26 — ATM_V2_P — JP27

Mains AC Power Energy Monitor | ATM90E36 | SPI DMA

[I2C]

U12 — DS3231SN — Address 0x68 — VCC, SCL, SDA, VBAT, INT/SQW, RST, 32KHZ, GND, NC

100n 10V — C31 — +3V3 — GND — +VBAT — RST_RTC — INT_RTC

CR2032 3V — J6 — Batt — C43 DNP — R48 5K1

RTC | DS3231SN | RTC Battery

2.5mm — H1 H2 H3 H4

Mounting

12V → +12V
+VBUS → +VBUS
+5V_ISO → +5V_ISO
3V3 +5V_ISO → +3V3

RTC_RST → RST_RTC
INT_RTC → INT_RTC
ADC_ALRT → ADC_ALRT
GP14 → GP14
ALM → TMP_ALM

CF1 → ATM_CF1
CF2 → ATM_CF2
CF3 → ATM_CF3
CF4 → ATM_CF4
ZX1 → ATM_ZX0
ZX2 → ATM_ZX1
ZX3 → ATM_ZX2
RST → RST_ATM
IRQ1 → ATM_IRQ0
IRQ2 → ATM_IRQ1
VP1 → ATM_V1_P
VP2 → ATM_V2_P
VP3 → ATM_V3_P
WO → ATM_WO

Test Points | Breakouts

ATM Status LEDs — ATM_CF1 R70 5K1 D5 Red; ATM_CF2 R71 5K1 D6 Green; ATM_CF3 R72 5K1 D7 Blue; ATM_CF4 R73 5K1 D8 Yellow

LED ATM90E36 Status

TXEN R74 23K7 D9 Blue
GPIO14 +3V3 R75 23K7 D10 Red GP14
OPTO_OUT R76 23K7 D11 Red
GPO_PWM R77 23K7 D12 Red
3V3 +3V3 R78 23K7 D13 Green
5V ISO +5V_ISO R79 23K7 D14 Green
12V +12V R80 124K D15 Green
VBUS +VBUS R81 23K7 D16 Green

LED General

Low Voltage AC Input for sensing mains volage and power direction flow.

AC_V1 — JP30 — R85 100K — 1K0 R86 — ATM_V1_P
AC_V2 — JP31 — R87 100K — 1K0 R88 — ATM_V2_P
AC_V3 — JP32 — R89 100K — 1K0 R90 — ATM_V3_P

AC In 8-19V RMS

AC input is referenced to board ground

Bell Transformer 240V AC RMS In
~Tapping 4V, Typically ~ 6.4 AC RMS Out (Not Ideal)
~Tapping 8V, Typically ~ 12.8 AC RMS Out
~Tapping 12V, Typically ~ 19.2 AC RMS Out (Ideal)
This is also used for board calibrated values.

AC Low Voltage Sense Inputs

Bell Transformer 240V AC RMS In
~Tapping 4V, Typically ~ 6.4 AC RMS Out (Not Ideal)
~Tapping 8V, Typically ~ 12.8 AC RMS Out
~Tapping 12V, Typically ~ 19.2 AC RMS Out (Ideal)
This is also used for board calibrated values.

DC or AC In 5-20V RMS — AC_DC_IN — D17 RB168VAM150 — D18 RB168VAM150 — JP32 — AC_V1

U14 — AP63203 – Fixed 3V3 Max 35V Input — VIN, BST, EN, SW, GND, FB — C62 100n 100V — L1 6u8 2.4A — C69 220u 35V — C70 100n 100V — C71 10u 35V — C72 Decoupling depending SMPS 100n 100V — C73 220u 35V — C74 100n 100V

Power Suppy SMPS | AC>DC Buck 3V3

U15 — LM2775DSGR — VIN, VOUT, PFM, C1+, OUTDIS, C1-, EN, GND-TP — C65 4u7 6V3 — C66 100n 10V — C63 4u7 6V3 — C67 22u 6V3 — C68 100n 10V

Power Suppy SMPS | DC Buck 5V ISO

[RGB]

D19 — WS2812B — LED_RGB — DI, VDD, DO, GND — C64 4u7 6V3

RGB | WS2812 | Status LED

IPEM Plus is an IoT Power Energy Monitor.
IPEM Plus is an ESP32-C5 SX1262 LoRa variant of the IPEM and EPEM series.

For further information and code examples, please see our code and GitHub WIKI pages. ditronix.net | GitHub.com/DitroniX | Hackster.io/DitroniX

Components optimised for BOM

DitroniX — Home Automation | Internet of Things | Energy Monitoring

Production

(C)2018-2026 Dave Williams | DitroniX | G8PUO

Enclosure VG-GR34 90x71x57
Main PCB 86 x 66.8

STEM

ESP32-C5 SX1262 SDK ditronix.net DitroniX Ltd – UK	IPEM Plus ESP32-C5 SX1262 E22-900M22S LoRa 1.2605.103	
	ATM90E36-ESP32-C5-LORA-IOT-POWER-ENERGY-MONITOR-BOARD-(PG2)	
	Sheet: 2/2	Drwg/Chk: Dave Williams
		06/05/2026 07:32